Assignment 9: Hash Tables
Software Engineering for Scientists

Objectives: Become familiar with hash tables as a means of integrating data.

Tasks:

1. This assignment will build off of assignment 8, so please use that repo.
2. Using test driven development
    a. Develop your own hash table data structure that can insert and retrieve key value pairs
        i. Write your own hashing function
        ii. Write your own collision handling methods
        iii. YOU DO NOT HAVE TO IMPLEMENT rehashing.
    b. Update your `get_data` function to optionally retrieve rows from a data file and return them as hash tables where the keys are the header row names
    c. Update `get_fire_gdp_year_data` and other functions to optionally use this new data structure.
        i. HINT do not replace your parallel arrays. You will need both data structures for your timing experiment.
3. Add internal timing using standard python functions to measure the runtime of creating and using parallel arrays and hash tables.
    a. Refer to the course notes for a full explanation https://github.com/swe4s/lectures/blob/main/doc/Profiling%20and%20Benchmarking.pdf
4. Add a section to your `README` that
    a. Describe your hash table
    b. Validated that your results with this new data structure match those that used search.
    c. Report on any speed improvements by using a hash table.
5. Create a release from master tagged as 3.0